In today's digital era, information has become one of the most valuable resources for individuals and organizations. Every successful decision, plan, or innovation depends on the ability to collect, process, and use information effectively. **Information Systems** play a vital role in making this possible.

This course introduces learners to the fundamental concepts of **Information Systems**, emphasizing how systems are analyzed, designed, and implemented to solve real-world problems. It explores how data flows within organizations, how systems support management at various levels, and how technology integrates with business processes to improve efficiency and decision-making.

Through this study, learners will gain an understanding of the **systematic approach to problem-solving**, including **System Analysis and Design (SAD)** the backbone of building reliable, efficient, and user-centered information systems. By the end of the course, students will be equipped with the knowledge to not only understand existing systems but also to participate in creating new ones that meet organizational needs.

Learning outcomes

By the end of this lesson, students should be able to

- Understand information system, describe its components and various types of IS.
- Explain the different phases of system development life cycle
- Understand and describe Phases of system and software Design

INTRODUCTION

When talking about systems, it is important to note that it is a combination of several components. A system is a group of interconnected parts, people, or processes that work together as a whole to achieve a common purpose. Key characteristics of a system include interdependent components, a defined purpose, and a unified whole where the performance of one part affects others. Systems can be tangible, like a computer, or abstract, like a mathematical formula.

An information system (IS) is an organized set of interconnected components such as hardware, software, data, people, and processes that work together to collect, process, store, and distribute information to support decision-making and organizational goals. These systems are essential for businesses and organizations to manage operations, interact with customers, and gain a competitive advantage by improving efficiency, productivity, and strategic planning in a rapidly evolving digital landscape.

Let's start with some basic ideas.

What is data?

Data is raw, unorganized facts that need to be processed to make it meaningful. Data can be something simple and seemingly random and useless until it is organized. Data does not have any specific purpose. An example of data can be the name of a person, a town, a figure, a number. If we just say 100, it makes no sense until we can associate that 100 with something and make meaning with it. If someone walks up to you and says, $10,000, you will look at that person as mad, until maybe they tell you the reason they mentioned such an amount of money.

What is Information?

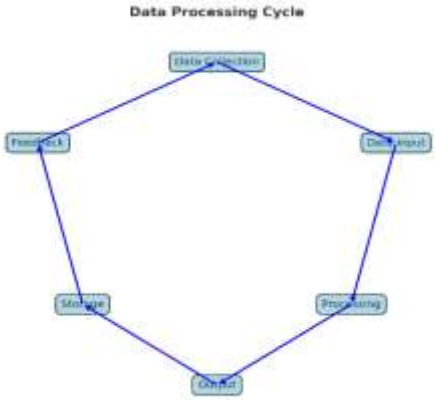
Information is data that has been organized, processed, interpreted and presented to be meaningful and useful. Putting that $10,000 we mentioned earlier into something useful will make sense and therefore will be considered as information.

Let us look at Information or data processing. This is the process that gives data meaning and transforms it into information.

The Data Processing Cycle

The **Data Processing Cycle** represents the continuous sequence of steps through which raw data is collected, transformed, and converted into meaningful information for decision-making. It

ensures that data is systematically handled from its origin to its final use. The cycle typically involves six key stages: **data collection, data input, processing, output, storage,** and **feedback.**



Here's the **Data Processing Cycle** diagram showing how information flows through the stages of:

1. **Data Collection**:

This is the first stage where raw data is gathered from various sources such as surveys, sensors, transactions, or records. The accuracy of this stage determines the quality of all subsequent results. For example, a sales company may collect data from customer invoices, receipts, or online orders to understand its total sales volume. Effective data collection must ensure completeness and relevance so that no essential information is missing.

2. **Data Input**:

Once the data has been collected, it must be entered into the system for processing. This stage involves data entry and data validation to ensure accuracy. Input devices such as keyboards, scanners, barcode readers, and mobile forms are often used. For instance, in a supermarket, barcodes are scanned to input product details and prices into the system. Validation checks, such as verifying product codes and quantities, prevent errors during this stage.

Data validation refers to the process of checking the data to ensure it is error free. It uses routines also called validation routines or check routines, that check for correctness, meaningfulness and security of data that are input to the system.

- ***format check***: e.g. date must be in the format DD/MM/YY
- ***Data type check***: e.g. the letter 'O' should not be typed instead of the integer 0.
- ***Range check***: The month should lie between 1 and 12
- ***Limit check***: the age should not be negative

- ***Presence check***: e.g. Any customer must have a telephone
- ***Consistency check***: e.g. If Title = "Mrs", then the gender = "F"

3. Data processing:

At this stage, the raw data is converted into meaningful information through several techniques such as classification, sorting, calculation, summarization, and comparison. For example, a payroll system processes employees' attendance and pay rate data to calculate their total salaries. The processing phase may involve manual computation or automated algorithms depending on the system's complexity. This is where the system performs its "thinking" function, transforming data into usable output. Some methods of processing include:

Batch Processing is a technique where data is collected over a period and then processed all at once as a single unit. For example, banks often process customers' salary payments at the end of the month in batches rather than individually, so that thousands of transactions are handled together.

Real-time Processing involves handling data immediately as it is entered into the system, giving instant results. A good example is at an ATM machine, where the balance of an account is updated instantly after cash is withdrawn.

Online Processing is like real-time but emphasizes continuous interaction between user and system through a network. An example is when students register for courses online, and the system immediately reflects available or occupied slots as they make selections.

Distributed Processing occurs when data processing tasks are divided among multiple computers connected in a network, often in different locations. A company like Google uses distributed processing to handle billions of search queries by spreading them across multiple servers worldwide.

Parallel Processing is when a task is split into smaller subtasks and processed simultaneously using multiple processors to save time. For example, in weather forecasting, huge volumes of climate data are analyzed in parallel by supercomputers to generate predictions faster.

Multiprocessing is related to parallel processing but takes place within a single computer that has more than one processor, allowing it to perform several processes at the same time. For instance, a modern laptop running antivirus scans in the background while the user streams a video, is using multiprocessing.

Time-sharing Processing allows multiple users to access a single system simultaneously, each getting a "slice" of the processing time. A university mainframe computer, for example, might allow hundreds of students to log in at once to check their exam results.

4. **Output**

After processing, the system produces results in a format that can be easily understood and used by decision-makers. The output may take the form of reports, visual dashboards, printouts, or digital summaries. For instance, a company's accounting software may produce a monthly financial report showing total revenue, expenses, and profit. This output serves as a key resource for planning and performance review.

5. **Storage**

This stage involves saving the processed data and results for future reference or further processing. Storage can be temporary (in memory) or permanent (in databases, hard drives, or cloud systems). For example, student records stored in a university database can later be retrieved for generating transcripts or analyzing performance trends. Proper storage also allows data security, backup, and retrieval.

6. **Feedback**

Feedback closes the loop in the data processing cycle. It involves reviewing the system's output and performance to identify errors, inefficiencies, or areas for improvement. For example, if a company realizes that sales data reports are missing certain product categories, feedback will trigger a revision in the data collection or input stages. This ensures continuous refinement and accuracy throughout the entire cycle.

We have talk about data validation. Let's talk about the verification.

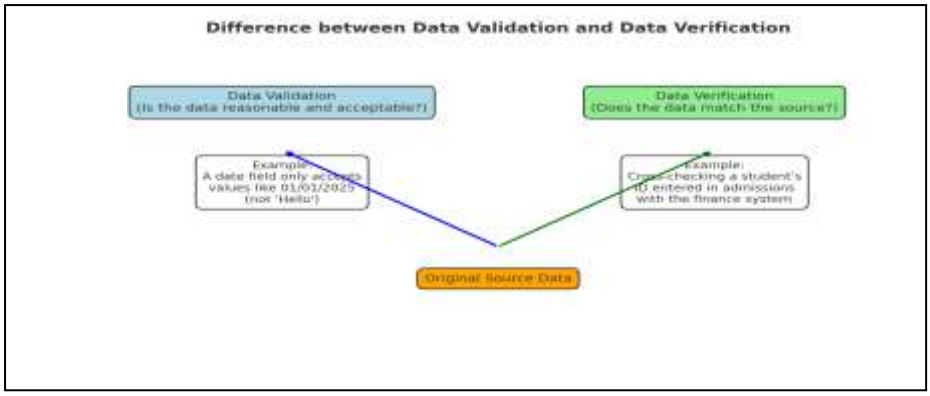
Data Verification is the process of ensuring that the data entered into a system matches the original source and is correctly recorded. While validation checks for accuracy and reasonableness, verification focuses on confirming that the data has been faithfully transferred from one place to another.

Double-entry Verification is a stage where data is entered twice, often by two different people or systems, and then compared. If the two entries do not match, the system highlights an error. For example, in accounting systems, transactions are often entered twice once as a debit and once as a credit to ensure the books balance.

Proofreading Verification involves carefully reviewing the data entered against the original source to spot any errors. This is common in publishing, where a proofreader checks typed text against handwritten manuscripts to ensure no mistakes were introduced during entry.

Cross Verification happens when data entered in one system is checked against data in another related system for consistency. For example, a student's registration number entered into the admissions database is cross-checked against the finance database to make sure the same student has also paid their fees.

Visual Verification is a stage where the user immediately checks the data on screen before submitting it, ensuring that it appears correct. For instance, when filling out an online form, the system often shows a summary page for the user to review before final submission.



Classification of Information

Information classification is a process in which organizations assess the data that they hold and the level of protection it should be given.

Organizations usually classify information in terms of

- confidentiality – i.e. who is granted access to see it. A typical system will include four levels of confidentiality:
    - Confidential (only senior management have access) • • •
    - Restricted (most employees have access)
    - Internal (all employees have access)
    - Public (everyone has access)

As you might expect, larger and more complex organizations will need more levels. Take hospitals, for example: doctors and nurses need access to patients' medical histories, which are highly

sensitive, but they shouldn't have access to other types of information that wouldn't fit that criteria, such as the hospital's financial records.

INFORMATION SYSTEM

What is a System? We can define a System as a combination of resources or functional units working together to accomplish a given task.

a) Basic element of a system. The basic elements of the system may be listed as:

→ Resources: Resources can be hardware, software or liveware (computer users).

→ Procedures: Every system functions under a set of rules that govern the system to accomplish the defined goal of the system. This set of rules defines the procedures for the system

→ Data/Information,

→ Processes: The systems have some processes that make use of the resources to achieve the set goal under the defined procedures.

b) Characteristics of a system

Systems also exhibit certain features and characteristics, some of which are:

→ **Objective**: Every system has a predefined goal or objective towards which it works. A system cannot exist without a defined objective.

→ **Standards**: It is the acceptable level of performance for any system. Systems should be designed to meet standards.

→ **Environment**: Every system whether it is natural or man-made co-exists with an environment. It is very important for a system to adapt itself to its environment.

→ **Feedback**: Feedback is an important element of systems. The output of a system needs to be observed and feedback from the output taken to improve the system and make it achieve the laid standards.

→ **Boundaries and interfaces**: Every system has defined boundaries within which it operates. Beyond these limits the system must interact with the other systems.

c) **Types of Systems**

→ **Manual Systems**: These are systems that are basically mechanical in nature and do not require or use any automation. Such systems existed predominantly in the past and were not very reliable or fast.

→ **Information Systems**: (IS) Any system that makes use of automation and especially harnessed by people, procedures, software, hardware and digital data is called an information system (IS). **What is an Information System**? An Information System (IS) is defined as an organized collection of technical and human resources for the processing, storing and distribution of an organization's information. An IS using a computer is called a Computer Based Information System (CBIS)

Components of IS

The 5 components that must come together to produce a Computer-Based Information system are:

- Hardware: The term hardware refers to machinery. This category includes the computer itself, which is often referred to as the central processing unit (CPU), and all its support equipment.
- Software: The term software refers to computer programs and the manuals (if any) that support them.
- Data: Data are facts that are used by programs to produce useful information.
- Procedures: Procedures are the policies that govern the operation of a computer system. "Procedures are to people what software is to hardware" is a common analogy that is used to illustrate the role of procedures in a CBIS.
- People: Often the most over-looked element of the CBIS are the people, probably the component that most influence the success or failure of information systems.

Importance of an IS in an organization

To gain the maximum benefits from your company's information system, you must exploit all its capacities. Importance of IS within an organisation are numerous:

→ Communication: Part of management is gathering and distributing information, and information systems can make this process more efficient by allowing managers to communicate rapidly.

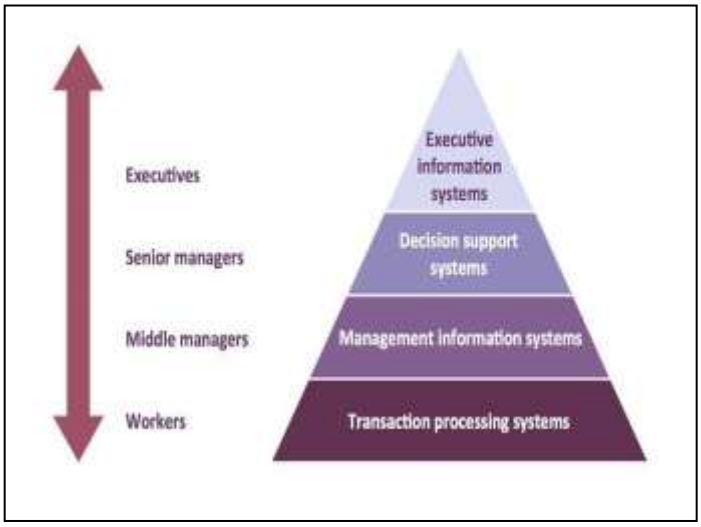
→ Operations: How you manage your company's operations depends on the information you have. Information systems can offer more complete and more recent information, allowing you to operate your company more efficiently.

→ Decisions: The company information system can help you make better decisions by delivering all the information you need and by modelling the results of your decisions.

→ Records: Your Company needs records of its activities for financial and regulatory purposes as well as for finding the causes of problems and taking corrective action.

Types of Information System Information systems differ in their business needs. Also depending upon different levels in organization information systems differ. Three major information systems are:

1. Transaction processing
2. Management information system
3. Decision support system
4. Executive Information System



Executive Information System (EIS):

This is a computer-based system intended to help and support the information and decision-making needs of senior executives of a company or organization. It provides easy access to internal and external information needed to achieve the strategic goals of the organization. It is commonly considered as a specialized form of DSS

Decision support system (DSS)

A decision support system is a computer application that helps users analyze problems and make business decisions more confidently. It uses data routinely collected in organizations and special analysis tools to provide information support to complex decisions. To store all the necessary decision-making data, DSSs or EISs often use extremely large databases, called data warehouses. A data warehouse stores and manages the data required to analyze historical and current business

circumstances. DSS manipulates and build upon the information from a MIS and/or TPS to generate insights and new information

Management Information system (MIS)

The MIS is defined as a system which provides information support for decision making in the organization. They use the results of transaction processing and some other information also. It is an information system that generates accurate, timely and organized information so managers and other users can make decisions, solve problems, supervise activities, and track progress. An important element of MIS system is database. MIS are built on the data provided by the TPS

Transaction Processing System (TPS)

A transaction processing system (TPS) is an information system that captures and processes data generated during an organization's day-to-day transactions. A transaction is any activity of an organization including a business activity such as a deposit, payment, order or reservation. As computers became more powerful, system developers built online transaction processing systems. With online transaction processing (OLTP) the computer processes transactions as they are entered.

Other types of information systems

Lists Detail reports Action reports Summary reports?

a) Office Automation System (OAS): These are systems that use varied computer machinery and software to digitally create, collect, store, manipulate and relay office information needed for accomplishing basic tasks in an organisation. All office functions like dictation, typing, filing, copying, fax, telex, microfilm and records management, telephone and telephone switch board operations are included.

b) Expert system (ES): An expert system is an information system that captures and stores the knowledge of human experts and then imitates human reasoning and decision-making processes for those who have less expertise. Expert systems are composed of two main components: a knowledge base and inference rules.

- A knowledge base is the combined subject knowledge and experiences of the human experts. The inference rules are a set of logical judgments applied to the knowledge base each time a user describes a situation to the expert system.

c) Integrated Information Systems (IIS):

With today's sophisticated hardware, software and communications technologies, it is often difficult to classify a system as belonging uniquely to one of the information system types discussed. Although expert systems still operate primarily as separate systems, organizations increasingly are consolidating their information needs into a single, integrated information system

Importance of an Information System

Information system has the following advantages to any organization or system:

a) IS helps an organization in decision making

b) They help to improve performance

c) Efficient in business communication

d) They help in record taking in an organization

e) Problem identification is easier with an IS in an organization.

Opportunities Provided By an Information System

Various career opportunities exist in the field of Information Systems. Thus, students can find an area that interests them and pursue it further. Although career opportunities are numerous, each one requires that a graduate possesses the following basic skills:

- Analytical skills
- Interpersonal skills
- Presentation skills
- Problem identification skills
- Report writing skills
- Self-development skills
- Time management skills

Career opportunities in the field of Information Systems include:

- Systems or business analyst
- Project manager Information officer
- Database administrator
- Business intelligence
- ICT consultant
- Web design
- Programmer

- Systems architect

ORGANIZATIONAL STRUCTURE

What's an organizational structure? An organizational structure defines how activities such as task allocation, coordination and supervision are directed towards the achievement of organizational aims. It can also be considered as the viewing glass or perspective through which individuals see their organization and its environment. Organizational structure determines how the roles, power and responsibilities are assigned, controlled, and coordinated, and how information flows between the different levels of management. A structure depends on the organization's objectives and strategy.

→ In a centralized structure, the top layer of management has most of the decision-making power and has tight control over departments and divisions.

→ In a decentralized structure, the decision making power is distributed and the departments and divisions may have different degrees of independence.

Why should you develop a structure for your organization?

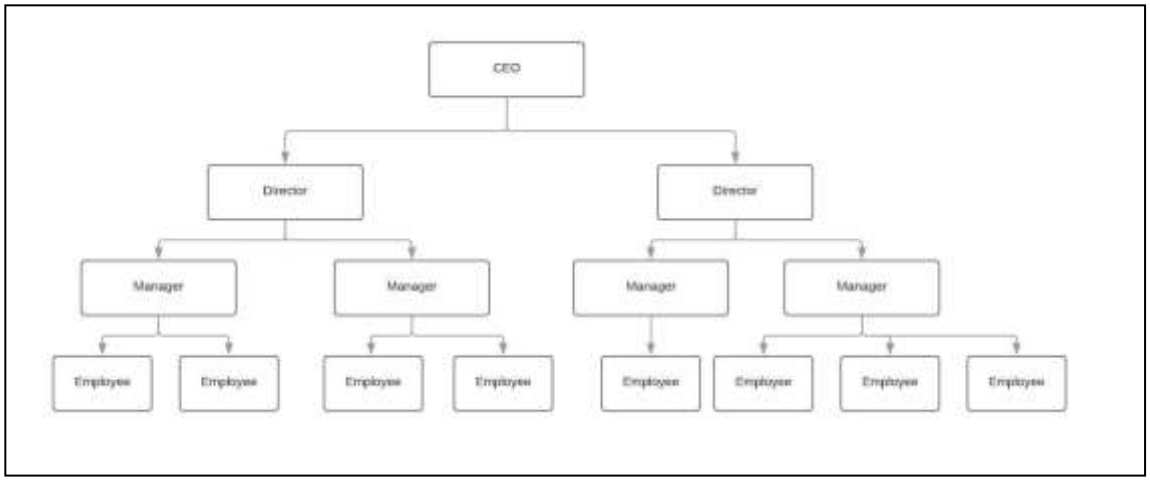
→ Structure gives members clear guidelines for how to proceed. A clearly established structure gives the group a means to maintain order and resolve disagreements.

→ Structure binds members together. It gives meaning and identity to the people who join the group, as well as to the group itself.

→ Structure in any organization is inevitable. An organization, by definition, implies a structure.

Organizational structure of an Information system

An information system's organizational structure refers to the formal arrangement of people, tasks, and resources within an information technology (IT) department, defining how technology functions are coordinated to support overall business goals. Key elements include a hierarchy or modular design, clearly assigned roles and responsibilities, reporting lines, and the coordination of IT activities (like networking, security, and support) to align with strategic objectives and foster efficient operations.


CEO
Director
Director
Manager
Manager
Manager
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Employee
Employee
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